

Probabilistic programming – Matus Course

Notes

This repo contains:

- [notes/raw-notes](#) – my raw notes
 - [notes/final-notes](#) – finalized handwritten notes packed in one PDF
 - [final-project/](#) – final project solution
 - [lectures-exercises/](#) – teacher notebooks with lecture presentations and exercises
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Notes – Terms by Lecture

Missing:

- Lecture9 (guest lecture: low priority)
- Lecture13 (cant find it)

L1 – Bayesian Modeling and Inference

1. Parameter
 2. Prior
 3. Likelihood
 4. Posterior
 5. Sample space
 6. Random variable
 7. Event
 8. Conditional probability
 9. Chain rule
 10. Law of total probability
 11. PMF: Probability mass function
 12. PDF: Probability density function
 13. CDF: Cumulative distribution function
 14. Grid approximation
 15. MCMC (basic part1)
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L2 – Regression

1. Bayesian linear regression vs linear regression
 2. Prior predictive check
 3. Posterior predictive check
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L3 – Multiple regression, confounds, categorical variables

1. Multiple regression
 2. Correlation is not causation
 3. Confounding
 4. Masking
 5. Data standardization
 6. Normalization
 7. PDP: Partial dependencies plots (residual plots)
 8. Causal inference (DAG)
 9. Counterfactual simulation
 10. Markov equivalent set
 11. Contrast plots
 12. Forest plot
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L4 – Multicollinearity, post-treatment bias, collider bias

1. Log-normal distribution
 2. Multi collinearity
 3. Post-treatment bias
 4. Collider bias
 5. Survivor bias
 6. Fork, Pipe, Collider, Descent
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L5 – Over/under-fitting and Interactions

1. Underfitting
2. Overfitting
3. Entropy $H(p)$

4. KL Divergencia
 5. LPPD – Log Pointwise Predictive Density
 6. Deviance
 7. Cross-Validation (CV)
 8. LOO – Leave One Out
 9. WAIC – Widely Applicable Information Criterion
 10. Regularization
 11. Robust Regression
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L6 – Interactions between regressors

1. Interaction
 2. Conditioning
 3. Index model
 4. Interaction term
 5. Symmetry of interactions
 6. Centering
 7. Survivorship bias
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L7 – Markov Chain Monte Carlo

1. MCMC (Markov Chain Monte Carlo)
 2. Metropolis algoritmus
 3. Proposal distribution
 4. Burn-in / Warm-up
 5. Autocorrelation
 6. ESS (Effective Sample Size)
 7. Gibbs algoritmus
 8. Conjugate prior
 9. HMC (Hamiltonian Monte Carlo)
 10. NUTS (No-U-Turn Sampler)
 11. Concentration of measure
 12. Diagnostika – Trace plot
 13. R-hat
 14. MCSE (Monte Carlo Standard Error)
 15. Divergences
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L8 – Generalized Linear Model (GLM)

1. Maximum Entropy princíp
 2. Maximum Entropy distribúcia
 3. GLM (Generalized Linear Model)
 4. Link function
 5. Logit link (logistická regresia)
 6. Log link
 7. Exponenciálna rodina distribúcií
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L9 – Variational Inference

Skip zatiaľ - guest lecture

L10 – Binomial, Poisson and multinomial regression

1. Binomial regression
 2. Aggregated vs disaggregated binomial
 3. Simpson's paradox
 4. Poisson regression
 5. Offset v Poisson regresii
 6. Multinomial regression
 7. Softmax
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L11 – Over-dispersion, zero-inflated outcomes, ordered categorical models

1. Over-dispersion
 2. Beta-Binomial
 3. Gamma-Poisson
 4. Zero-inflated model
 5. Ordinal regression
 6. Cumulative log-odds
 7. Ordered predictor
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L12 – Multilevel models

1. Multilevel model
 2. Pooling
 3. Shrinkage
 4. More than one cluster
 5. Centered vs Non-centered model
 6. Divergent transitions
 7. Posterior predictions v multilevel modeloch
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L13 – Bayesian Inference for Privacy Risk Analysis + Exam publication

Neviem najst lecture notes